

The empirical recurrence for OEIS A316737, recovered from its tail

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8 September 2026

Abstract

OEIS A316737 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 160$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{93} = 36864 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A316737 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 1, 2, 3, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 70, 210, 410, 4956, 25047, 75856, 491644, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Jul 11 2018 20:32:33, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 160$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 160$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 320$ exact terms from $n = 5$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	3	c_{25}	115129086	c_{49}	−117635991981	c_{73}	−3706503513
c_2	8	c_{26}	913989214	c_{50}	98458686872	c_{74}	−1821101264
c_3	90	c_{27}	142420593	c_{51}	247415177407	c_{75}	148344099
c_4	−211	c_{28}	−893409950	c_{52}	173803024986	c_{76}	666998119
c_5	−832	c_{29}	624096294	c_{53}	−46746921098	c_{77}	467514632
c_6	−3018	c_{30}	4756047808	c_{54}	−202338153514	c_{78}	265456732
c_7	5844	c_{31}	5000032933	c_{55}	−163226547717	c_{79}	148491313
c_8	28280	c_{32}	−6109184713	c_{56}	985170421	c_{80}	5827742
c_9	57818	c_{33}	−19746454683	c_{57}	126420561543	c_{81}	−123088790
c_{10}	−80804	c_{34}	−18458972702	c_{58}	122482223170	c_{82}	−136753992
c_{11}	−496349	c_{35}	11426181017	c_{59}	24386490907	c_{83}	−60806456
c_{12}	−781016	c_{36}	44610778006	c_{60}	−67045792546	c_{84}	15765356
c_{13}	640555	c_{37}	45550629316	c_{61}	−84996932715	c_{85}	36666212
c_{14}	5388034	c_{38}	−2695090173	c_{62}	−33683385391	c_{86}	19494912
c_{15}	7738149	c_{39}	−59476276542	c_{63}	30610045855	c_{87}	1145648
c_{16}	−3247506	c_{40}	−71758667737	c_{64}	53422809476	c_{88}	−4716000
c_{17}	−38972078	c_{41}	−31940660659	c_{65}	27881027415	c_{89}	−2556688
c_{18}	−53321686	c_{42}	23717193612	c_{66}	−12046602580	c_{90}	−519712
c_{19}	11172033	c_{43}	58314526703	c_{67}	−27494379978	c_{91}	219840
c_{20}	189556497	c_{44}	83083036396	c_{68}	−14825098517	c_{92}	103680
c_{21}	237045087	c_{45}	81880677834	c_{69}	4868697757	c_{93}	36864
c_{22}	−27376143	c_{46}	17402401297	c_{70}	11367042928		
c_{23}	−587682365	c_{47}	−115525599561	c_{71}	5591574622		
c_{24}	−588727039	c_{48}	−201860879527	c_{72}	−1799565220		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A316737 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -36864$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 36864.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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